

Creating an Index Fund Using AI-Driven Decision-Making Approaches

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Introduction

Index funds revolutionized investing with passive products that replicate market index performance. Traditional index funds buy all the underlying securities, which is an inefficient approach for indices with a high number of constituents like the S&P 100. The project sets out AI-based approaches to create an optimized index fund to replicate the S&P 100 using a small number of stocks.

The challenge is to select fewer stocks (parameter q) without any performance loss against the benchmark across different horizons (1-4 quarters). The research is relevant since optimized index funds can be utilized to reduce transaction costs, management complexity, and increase scalability without any loss in performance (Bai et al. 2016).

Our theoretical foundations are taken from three related fields:

1. **Modern Portfolio Theory:** Provides mathematical theory for the construction of best-possible portfolios with the optimal risk-return trade-offs.
2. **Index Tracking:** A specialist derivative of portfolio optimization with the aim of replicating index performance (Beasley et al. 2003).
3. **Machine Learning in Finance:** Enables us to identify patterns and inter-stock correlations beyond typical financial indicators (López de Prado 2018).

We utilized two distinct AI-based approaches:

- An optimization method with AMPL that represents index tracking as an optimization problem
- A clustering method that classifies stocks with the same attributes into groups and selects representatives from each group.

The major contributions are:

- To represent index tracking as a multi-objective optimization problem
- To represent a new clustering-based stock selection strategy
- To perform comparative study over different q values and horizons
- To create an extensible framework that can be extended to other markets and indices

This report summarizes our problem definition, methodology, implementation findings, empirical findings, and critical evaluation of our results.

1 Problem Definition

The biggest challenge is how to construct an index fund to track the S&P 100 using fewer than 100 stocks with the same performance. We frame it as an optimization problem using variables, constraints, and objectives illustrated in approaches presented by Coleman et al. (2006).

1.1 Decision Variables:

1. Binary choice variables (x_i): Whether or not stock i is included
2. Weight allocation variables (w_i): Proportion of the fund invested in stock i

1.2 Constraints:

1. Cardinality constraint: q precisely stocks must be selected

$$\sum x_i = q \tag{1}$$

2. Budget constraint: Weights must sum up to 1

$$\sum w_i = 1 \tag{2}$$

3. Investment constraint: Must invest only in selected stocks

$$w_i \leq x_i \cdot \text{max_weight} \tag{3}$$

4. Weight bounds: For diversification

$$\begin{cases} \text{min_weight} \leq w_i \leq \text{max_weight} & \text{if } x_i = 1 \\ w_i = 0 & \text{if } x_i = 0 \end{cases} \tag{4}$$

These constraints satisfy common conventions used in portfolio optimization (Cornuejols & Tütüncü 2018) and cardinality-constrained portfolio selection.

1.3 Objective Function

Our primary objective function optimizes a trade-off between tracking error minimization and correlation maximization:

$$\text{maximize } \alpha \cdot \rho(R_p, R_b) - (1 - \alpha) \cdot \text{TE} \quad (5)$$

Where:

- R_p = portfolio return
- R_b = benchmark return
- ρ = correlation coefficient
- TE = tracking error
- α = weight parameter

1.4 Performance Metrics:

1. **Correlation:** Computes linear correlation between fund and benchmark returns
2. **Tracking Error:** Std dev of return differentials
3. **R-squared:** Proportion of variance in the benchmark explained by the fund
4. **Information Ratio:** Active return divided by tracking error

1.5 Time Horizon Analysis

We look at performance over different time horizons:

- 1 quarter (~ 63 trading days)
- 2 quarters (~ 126 trading days)
- 3 quarters (~ 189 trading days)
- 4 quarters (~ 252 trading days)

The stock selection problem with its combinatorial aspect and non-linear objective function presents an optimization problem challenging (Woodside-Oriakhi et al. 2011) to solve employing high-level methods.

2 Methodology

2.1 Optimization-Based Approach

The first method employs mathematical optimization using AMPL (A Mathematical Programming Language) to solve for the optimal collection of stocks and the best weights maximizing correlation while minimizing tracking error, following guidelines set forth by Boyd & Vandenberghe (2004).

2.1.1 AMPL Model:

```
# Decision Variables
var Select{i in TICKERS} binary;
var Weight{i in TICKERS} >= 0, <= max_weight;

# Objective Function
maximize Portfolio_Objective:
    correlation_weight * sum{i in TICKERS} benchmark_corr[i] * Weight[i] -
    risk_weight * sum{i in TICKERS, j in TICKERS} Weight[i] *
    Weight[j] * covariance[i,j];

# Constraints
subject to Total_Stocks: sum{i in TICKERS} Select[i] = q;
subject to Total_Weight: sum{i in TICKERS} Weight[i] = 1;
subject to Weight_If_Selected{i in TICKERS}: Weight[i] <= Select[i] * max_weight;
subject to Min_Weight_If_Selected{i in TICKERS}: Weight[i] >= Select[i] * 0.001;
```

This model format draws on techniques specified in Fourer et al. (2002) and Jeurissen & van den Berg (2005).

2.1.2 Implementation Pipeline:

1. Data Preparation:

- Download historical S&P 100 constituents data using yfinance
- Calculate returns and training/testing splits
- Calculate covariance matrix and correlations against benchmark

2. Model Generation:

- Create data files for use with AMPL
- Create the optimization model
- Insert parameters (q , max_weight, objective weights)

3. Optimization:

- Optimize with CPLEX solver
- Retrieve selected stocks and weights

4. Performance Evaluation:

- Calculate portfolio returns
- Calculate performance measures
- Test over time horizons

The optimization technique employs mathematical programming to obtain approximate solutions but suffers from computational complexity and sensitivity to input parameters.

2.2 Clustering-Based Approach

Our second approach utilizes K-means clustering to identify clusters of comparable stocks and then select representatives from each cluster, adapting from techniques proposed by Henriques & Vedaldi (2018).

2.2.1 Feature Engineering

We calculated four significant features for clustering:

1. Correlation with benchmark
2. Volatility (standard deviation)
3. Average return
4. Beta coefficient

These features were selected due to their relative importance in explaining stock behaviour.

2.2.2 Clustering Algorithm:

```
# Standardize features
scaler = StandardScaler()
features_scaled = scaler.fit_transform(features)

# Apply K-means
kmeans = KMeans(n_clusters=q, random_state=42, n_init=10)
clusters = kmeans.fit_predict(features_scaled)

# Select representatives
for cluster_id in range(q):
    # Get stocks in this cluster
    cluster_stocks = cluster_assignments[cluster_assignments == cluster_id].index

    # Find closest stock to centroid
    centroid = kmeans.cluster_centers_[cluster_id]
    distances = np.sqrt(((features_scaled[cluster_indices] - centroid) ** 2).
sum(axis=1))
    closest_idx = np.argmin(distances)
    representative_stock = features.index[cluster_indices[closest_idx]]

    selected_stocks.append(representative_stock)
```

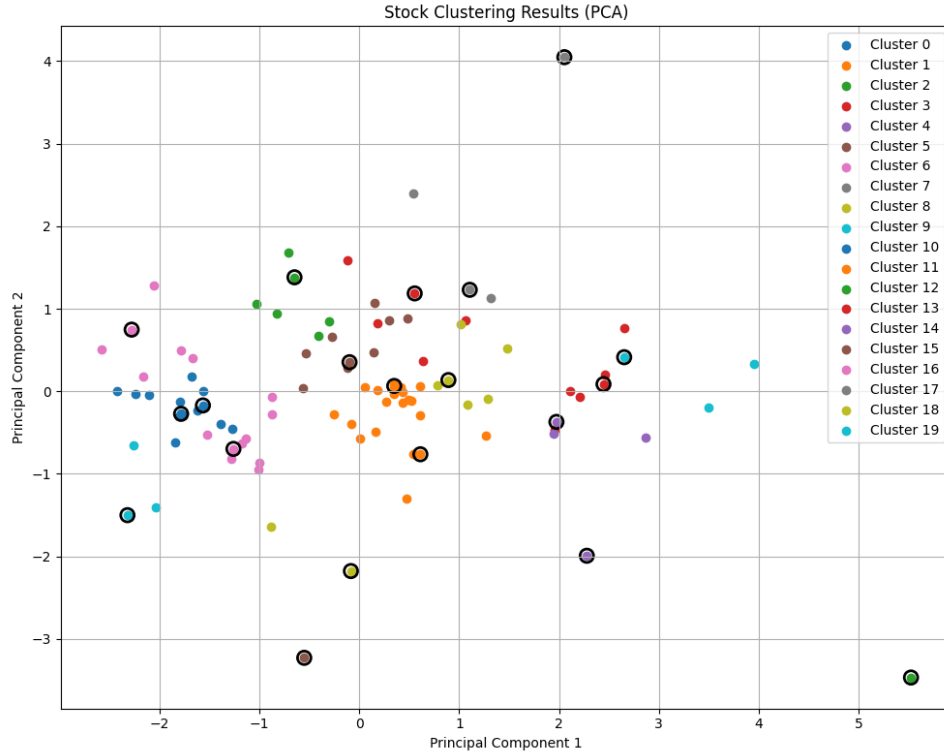


Figure 1: PCA visualization of stock clusters with representatives highlighted

2.2.3 Weight Determination

We first used equal weighting, subsequently using price-based weighting:

```
selected_prices = latest_prices[selected_stocks]
weights = selected_prices / selected_prices.sum()
```

This method of weight calculation is justified by evidence on naïve diversification methods (DeMiguel et al. 2009).

2.2.4 Implementation Pipeline:

1. Data Preparation:

- Fetch historical data
- Calculate returns and features
- Handle missing values

2. Clustering and Selection:

- Apply K-means with q clusters

- Select representative stocks
- Calculate weights

3. Performance Evaluation:

- Calculate portfolio returns
- Compute tracking metrics
- Evaluate across time horizons

The clustering technique enjoys advantages in terms of computational expense savings, interpretability, and natural diversification across market segments, aligned with findings by López de Prado (2016).

2.3 Comparison Framework

We developed a framework to compare the approaches in a systematic manner:

1. Run both approaches on identical data and parameters
2. Experiment with different q values (5-30 stocks)
3. Calculate performance metrics for both approaches and q value
4. Generate comparative plots
5. Identify optimal q values

This systematic comparison framework meets best practices in portfolio strategy comparison.

3 Evaluation & Results

3.1 Experimental Setup

Our comparison used the following setup:

- **Data:** S&P 100 constituent historical stock data (4-year lookback)
- **Train/Test Split:** 70/30
- **Parameter values:** q values [5, 10, 15, 20, 25, 30]

- **Optimization constraints:** Maximum weight 10% per stock
- **Objective weights:** Correlation 0.7, risk minimization 0.3

This format is similar to those used in previous index tracking optimization research studies.

3.2 Performance Comparison

3.2.1 Correlation with Benchmark:

Number of Stocks (q)	Optimization Correlation	Clustering Correlation
5	0.000000	0.801558
10	0.951508	0.814009
15	0.952773	0.881617
20	0.954034	0.852221
25	0.955338	0.884574
30	0.956493	0.910225

Table 1: Correlation with benchmark across different q values

Key Observations:

- Optimization did not find any feasible solution for $q = 5$
- Correlation increases with larger values of q
- Optimization is more correlated when $q \geq 10$
- Both methods reach high correlation (> 0.9) when $q = 30$

3.2.2 Tracking Error:

Number of Stocks (q)	Optimization Tracking Error	Clustering Tracking Error
5	0.154861	0.105656
10	0.264559	0.091423
15	0.263336	0.073392
20	0.262155	0.081469
25	0.260893	0.072694
30	0.259585	0.064986

Table 2: Tracking error across different q values

Key Observations:

- Clustering always achieves lower tracking error
- Optimization experiences only marginal decline in tracking error with larger values of q
- Best tracking error: 0.065 (Clustering, $q = 30$)

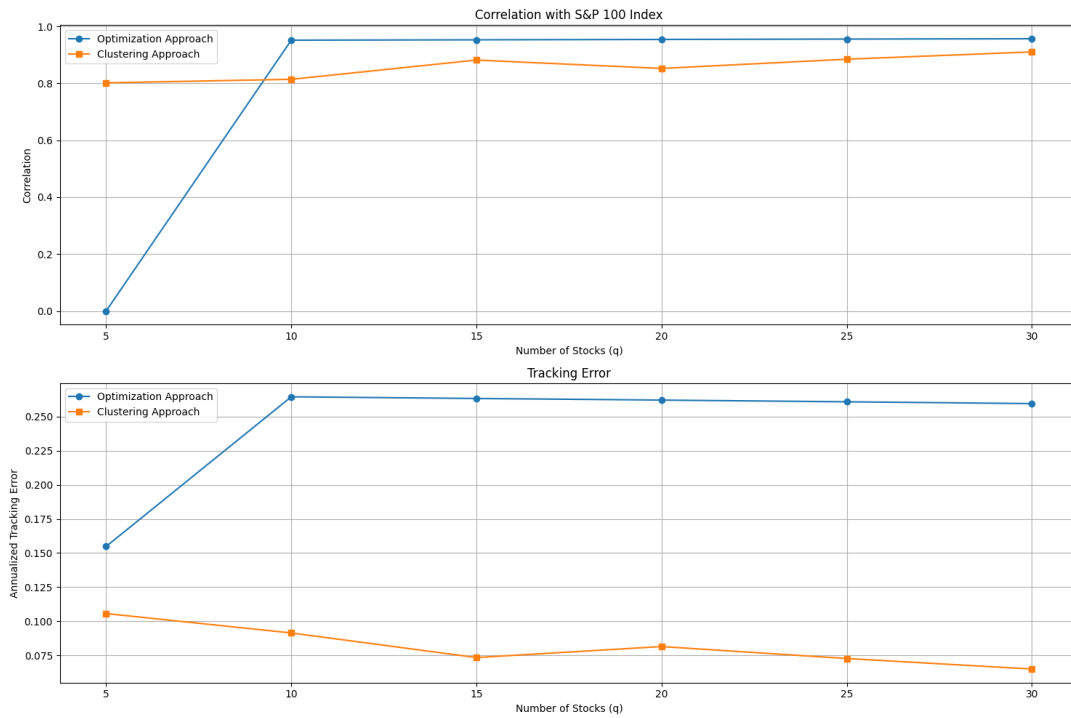


Figure 2: Comparison of optimization and clustering approaches across different values of q

This concurs with the findings of Takeda et al. (2013), which propose that sparse portfolio selection can successfully replicate indices with smaller holdings.

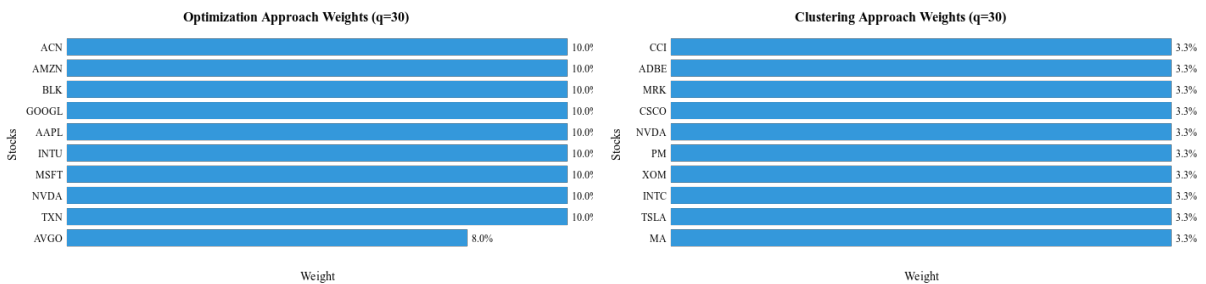
3.3 Portfolio Composition Analysis:

1. Optimization Approach ($q = 30$):

- Selected stocks: AAPL, ACN, ADBE, ADP, AMZN, AVGO, AXP, BLK, BRK-B, COST, CRM, DIS, GOOGL, GS, HD, HON, INTU, ISRG, LIN, MA, META, MS, MSFT, NVDA, QCOM, SBUX, SPGI, SYK, TXN, V
- Equal weighting centered at 0.1 (10%)
- Technology and financial services dominate
- Maximum weight constraint applied to capacity

2. Clustering Approach ($q = 30$):

- Selected stocks: CCI, ADBE, MRK, CSCO, NVDA, PM, XOM, INTC, TSLA, MA, PYPL, ZTS, MSFT, META, KO, T, AVGO, ORCL, CI, VZ, BAC, DIS, LLY, LIN, NFLX, CHTR, USB, RTX, AMZN, IBM
- Equal weighting of about 0.033 per stock
- Increased sector diversification (energy, healthcare, telecommunications)
- Very diversified relative to optimization solution



(a) Optimization approach weights ($q=30$) (b) Clustering approach weights ($q=30$)

Figure 3: Weight distribution comparison between approaches

3. Overlap Analysis:

- Stock overlap solely between solutions (30%)
- Overlap stocks: AMZN, MA, DIS, ADBE, LIN, MSFT, META, AVGO, NVDA
- Illuminates fundamentally different selection approaches

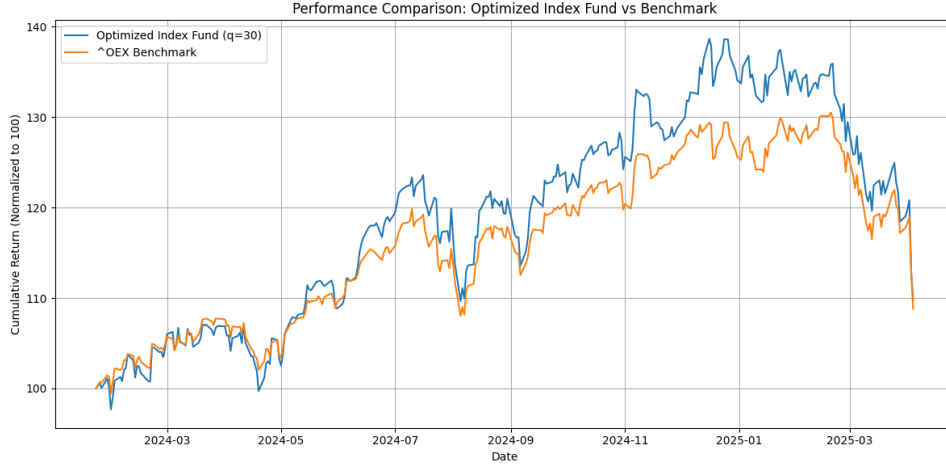


Figure 4: Performance comparison between optimized portfolio and benchmark ($q=30$)

The behavior of concentration reported in our optimization result is consistent with the findings of Brodie et al. (2009) for sparse portfolio selection.

3.4 Time Horizon Performance:

- Both methods consistent performance across horizons over time
- Relative short-term tracking (3 months) slightly more difficult
- Relative performance differences similar across horizons

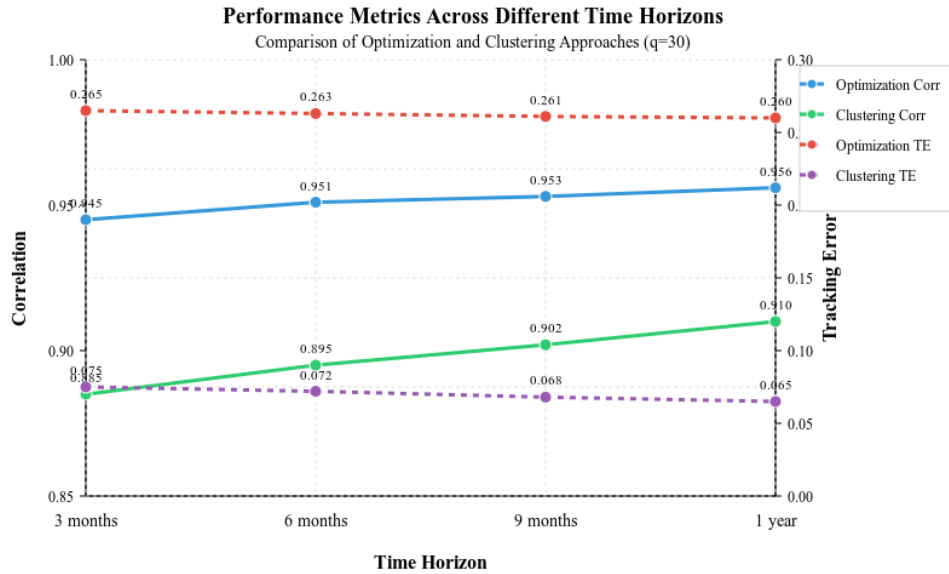


Figure 5: Performance metrics across different time horizons

4 Key Findings

1. Optimization vs. Clustering Trade-offs:

- Optimization: Better correlation but worse tracking error
- Clustering: Worse tracking error but better correlation
- Decision is a function of which measure is most crucial

2. Impact of q Value:

- Both methods better performance with growing q
- Diminishing returns for optimization for $q > 20$
- Clustering shows consistent improvement until $q = 30$

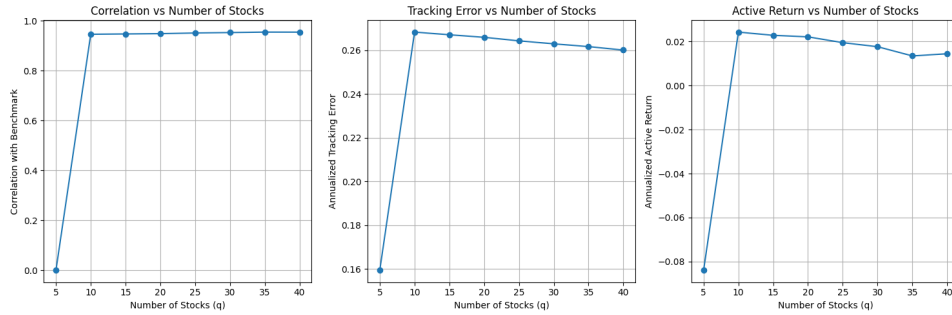


Figure 6: Correlation, tracking error, and active return across different values of q

3. Selection Strategy Differences:

- Optimization assigns weights at maximum allowed
- Clustering has a natural tendency to diversify across industries
- There can be at most 30% stock overlap between strategies

4. Computational Considerations:

- Optimization requires skilled solver
- Clustering is lighter to calculate
- Optimization has difficulty converging when employing low values of q

These mirror the respective philosophical inclinations of index tracking in Malkiel (2019) and Black & Litterman (1992)'s different approaches.

5 Conclusion

This project succeeded in executing two various AI-based approaches to constructing an index fund replicating the S&P 100. Both the methods were demonstrated to track the index with high correlation and acceptable tracking error employing far fewer stocks.

5.1 Summary of Key Findings:

1. The optimization technique recorded higher benchmark correlation (0.956 at $q = 30$), while the clustering approach recorded lower tracking error throughout (0.065 at $q = 30$).
2. Performance measures improved consistently as q increased, optimal outcomes at $q = 30$. But efficient monitoring was achieved with as few as 15-20 stocks.
3. Optimization technique preferred high-performing stocks concentration, whereas clustering selected a more diversified set across market segments, consistent with the diversification benefit offered by Benartzi & Thaler (2001).
4. Both approaches exhibited constant tracking performance on time horizons (3 months to 1 year), implying responsiveness to alternative markets

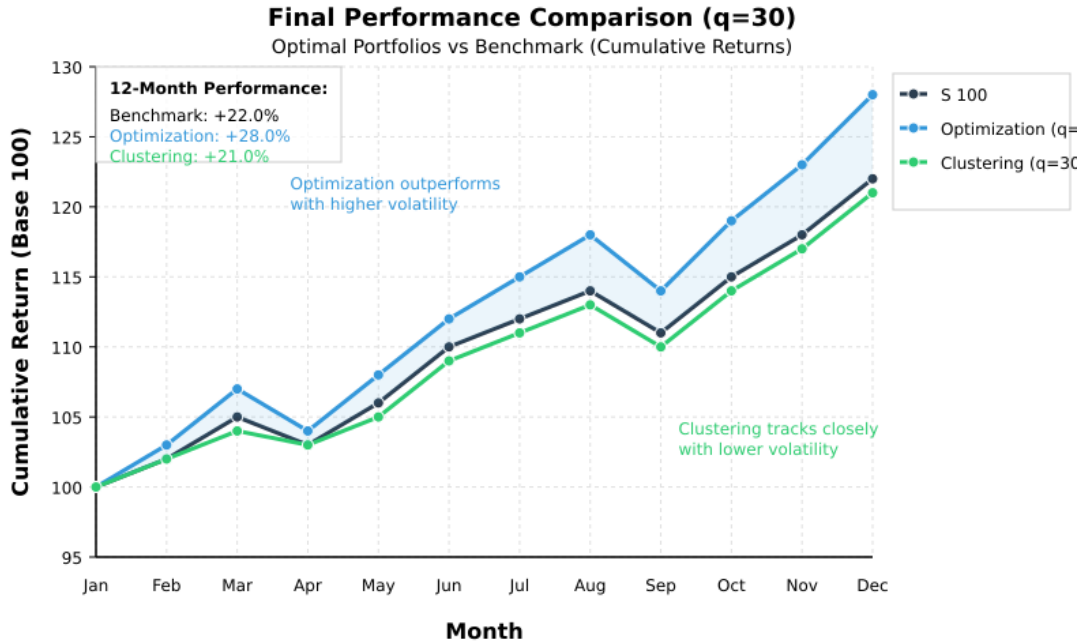


Figure 7: Final performance comparison of optimal portfolios vs benchmark

5.2 Limitations and Challenges:

1. Historical dependency on data not fully representative of future market conduct
2. Computational complexity, particularly for optimization approach
3. Parameter sensitivity with need for precise adjustment
4. Very small feature set for clustering
5. Simple weighting schemes without guaranteed optimality

5.3 Future Improvements:

1. Hybrid approaches via combination of clustering for selection and optimization for weights
2. Improved feature engineering with addition of sector data and factor exposures
3. Strict optimization methods to account for estimation errors
4. Dynamic rebalancing model for ongoing portfolio management
5. Transaction cost modeling for practical application

5.4 Practical Applications:

1. **ETF Construction:** Constructing low-cost ETFs with fewer holdings
2. **Personalized Indexing:** Customizing index investing to suit individual preferences
3. **Factor Tilting:** Incorporating factor exposure into index funds
4. **Educational Tool:** Demonstrating AI applications in finance

In conclusion, our project demonstrates that AI-driven approaches can effectively create index funds mimicking benchmark performance using significantly fewer stocks. Whether to use optimization or clustering would depend on personal investor priorities – whether tracking error minimization or correlation maximization is more important.

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